

Investigating a Lumma Stealer Detection Using Wireshark: A Step-by-Step Analysis

In this investigation, we analyse a packet capture (PCAP) file to identify a potentially compromised Windows system communicating with a Malware (Lumma Stealer). Using Wireshark, we systematically uncover key indicators such as the infected host's IP address, MAC address, hostname, and associated user details. The known malicious indicator used for this analysis is:

- Lumma Stealer IP Address: **153.92.1[.]49**

Lumma Stealer is an information-stealing malware family that specializes in collecting sensitive data from infected systems. Before stealing credentials or browser data, the malware often performs victim fingerprinting, gathering information about the operating system, hostname, user account, and environment. This information is transmitted to attacker-controlled infrastructure and helps threat actors determine the value of the compromised system.

Scenario Overview

The investigation begins with a known malicious network indicator:

- Lumma Stealer IP Address: **153.92.1[.]49**
- Port: 443
- Timestamp: 2026-01-27 at 23:05 UTC

Environment Details

- LAN Segment: 10.1.21.0/24
 - Gateway: 10.1.21[.]1
 - Broadcast: 10.1.21[.]255
 - Domain: win11office[.]com
 - Active Directory Domain: WIN11OFFICE
 - Domain Controller: 10.1.21[.]2 (WIN-LU4L24X3UB7)
-

What is the IP address of the infected Windows client?

To begin the investigation, the PCAP file was loaded into Wireshark and filtered using:

`"ip.addr==153.92.1.49"`

This filter isolates all traffic where the RAT IP appears as either the source or destination.

This will show us all the communication from the hosts which is communicating with **153.92.1[.]49**. We only see 10.1.21[.]58.

We can verify this by going into Statistics -> Conversations. The results confirmed that 100% of the filtered traffic involved communication exclusively between 10.1.21[.]58 and 153.92.1[.]49, strongly indicating that this internal host is the infected system.

Ethernet · 1	IPv4 · 1	IPv6	TCP · 24	UDP							
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	
10.1.21.58	153.92.1.49	2,968	2 MB	50	2,968	100.00%	1,739	2 MB	1,229	150 kB	

Infected Machine: **10.1.21[.]58**

What is the MAC address of the infected Windows client?

Using the same filtered dataset, the first packet was examined in detail. Within the Ethernet frame header, the source MAC address corresponding to the infected system was identified.

```
▶ Frame 23724: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
▶ Ethernet II, Src: Intel_c8:0e:f2 (00:21:5d:c8:0e:f2), Dst: Cisco_be:8c:d4 (00:04:c1:be:8c:d4)
▶ Internet Protocol Version 4, Src: 10.1.21.58, Dst: 153.92.1.49
▶ Transmission Control Protocol, Src Port: 56198, Dst Port: 443, Seq: 0, Len: 0
```

MAC_Address: **00:21:5d:c8:0e:f2**

What is the host name of the infected Windows client?

To identify the system hostname, DHCP traffic was analysed using the filter: `"dhcp"` DHCP Request packets often include the hostname of the client as part of the request. Among the filtered results, four DHCP packets were observed.

By correlating the MAC address (00:21:5d:c8:0e:f2) with the DHCP Request packet, the hostname associated with the infected system was identified as:

```
▶ Option: (50) Requested IP Address (10.1.21.58)
▼ Option: (12) Host Name
  Length: 15
  Host Name: DESKTOP-ES9F3ML
```

Hostname: **DESKTOP-ES9F3ML**

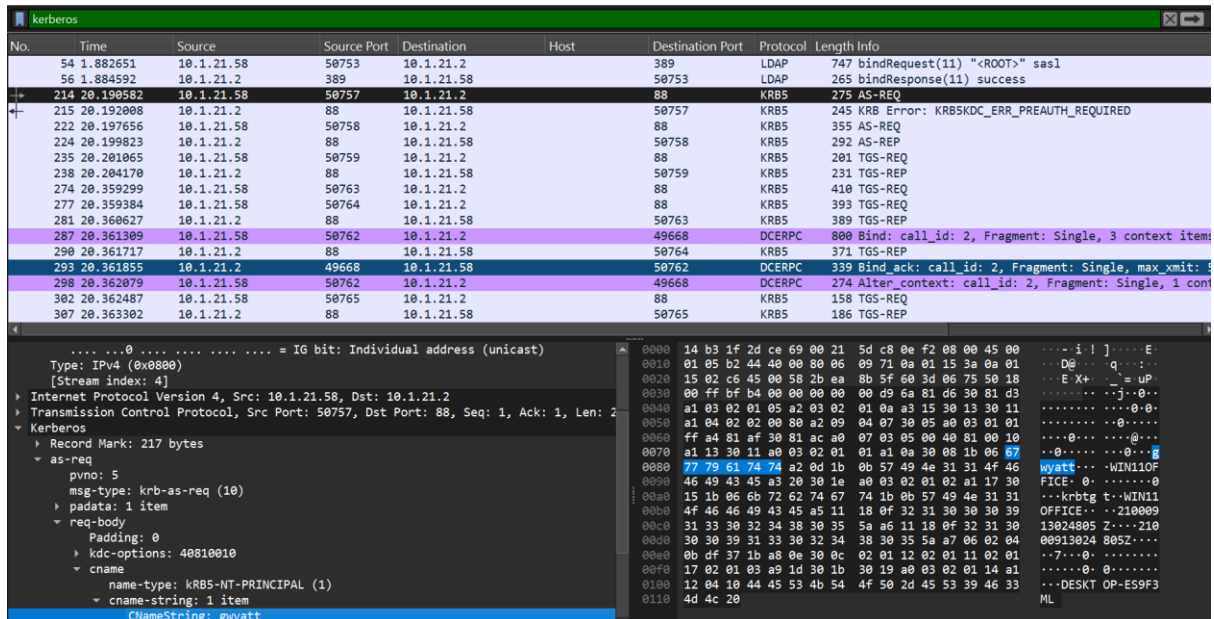
What is the user account name from the infected Windows client?

Next, Kerberos authentication traffic was analyzed to determine the user account associated with the infected host.

Applying the filter:

"kerberos"

allowed isolation of authentication-related packets.

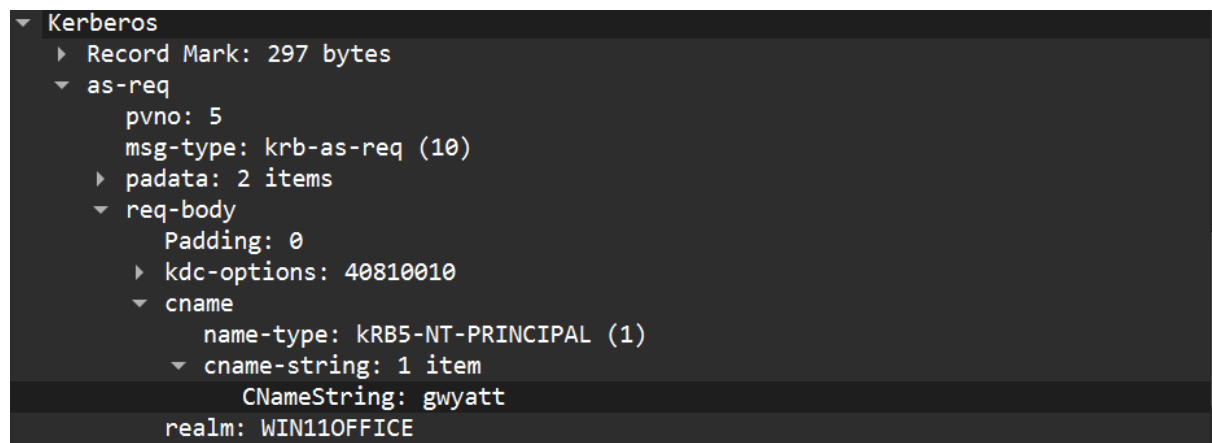


The image shows a Wireshark packet capture of Kerberos traffic. The top pane displays a list of packets, with packet 214 selected. The bottom pane shows the details of packet 214, which is a Kerberos AS-REQ message. The 'CNameString' field is highlighted, showing the value 'gwyatt'.

No.	Time	Source	Source Port	Destination	Host	Destination Port	Protocol	Length	Info
54	1.882651	10.1.21.58	50753	10.1.21.2		389	LDAP	747	bindRequest(11) "<root>" sasl
56	1.884592	10.1.21.2	389	10.1.21.58		50753	LDAP	265	bindResponse(11) success
214	20.190562	10.1.21.58	50757	10.1.21.2		88	KRB5	275	AS-REQ
215	20.192088	10.1.21.2	88	10.1.21.58		50757	KRB5	245	KRB Error: KRB5KDC_ERR_PREAUTH_REQUIRED
222	20.197656	10.1.21.58	50758	10.1.21.2		88	KRB5	355	AS-REQ
224	20.198023	10.1.21.2	88	10.1.21.58		50758	KRB5	292	AS-REP
235	20.201065	10.1.21.58	50759	10.1.21.2		88	KRB5	201	TGS-REQ
238	20.204170	10.1.21.2	88	10.1.21.58		50759	KRB5	231	TGS-REP
274	20.359299	10.1.21.58	50763	10.1.21.2		88	KRB5	410	TGS-REQ
277	20.359384	10.1.21.58	50764	10.1.21.2		88	KRB5	393	TGS-REQ
281	20.360627	10.1.21.2	88	10.1.21.58		50763	KRB5	389	TGS-REP
287	20.361309	10.1.21.58	50762	10.1.21.2		49668	DCERPC	800	Bind: call_id: 2, Fragment: Single, 3 context items
290	20.361717	10.1.21.2	88	10.1.21.58		50764	KRB5	371	TGS-REP
293	20.361855	10.1.21.2	49668	10.1.21.58		50762	DCERPC	339	Bind ack: call_id: 2, Fragment: Single, max xmit: 4
298	20.362079	10.1.21.58	50762	10.1.21.2		49668	DCERPC	274	Alter_context: call_id: 2, Fragment: Single, 1 context item
302	20.362487	10.1.21.58	50765	10.1.21.2		88	KRB5	158	TGS-REQ
307	20.363302	10.1.21.2	88	10.1.21.58		50765	KRB5	186	TGS-REP

Details of packet 214 (Kerberos AS-REQ):

- Type: IPv4 (0x0800)
- [Stream index: 4]
- Internet Protocol Version 4, Src: 10.1.21.58, Dst: 10.1.21.2
- Transmission Control Protocol, Src Port: 50757, Dst Port: 88, Seq: 1, Ack: 1, Len: 275
- Kerberos
 - Record Mark: 217 bytes
 - as-req
 - pvno: 5
 - msg-type: krb-as-req (10)
 - padata: 1 item
 - req-body
 - Padding: 0
 - kdc-options: 40810010
 - cname
 - name-type: KRB5-NT-PRINCIPAL (1)
 - cname-string: 1 item
 - CNameString: gwyatt



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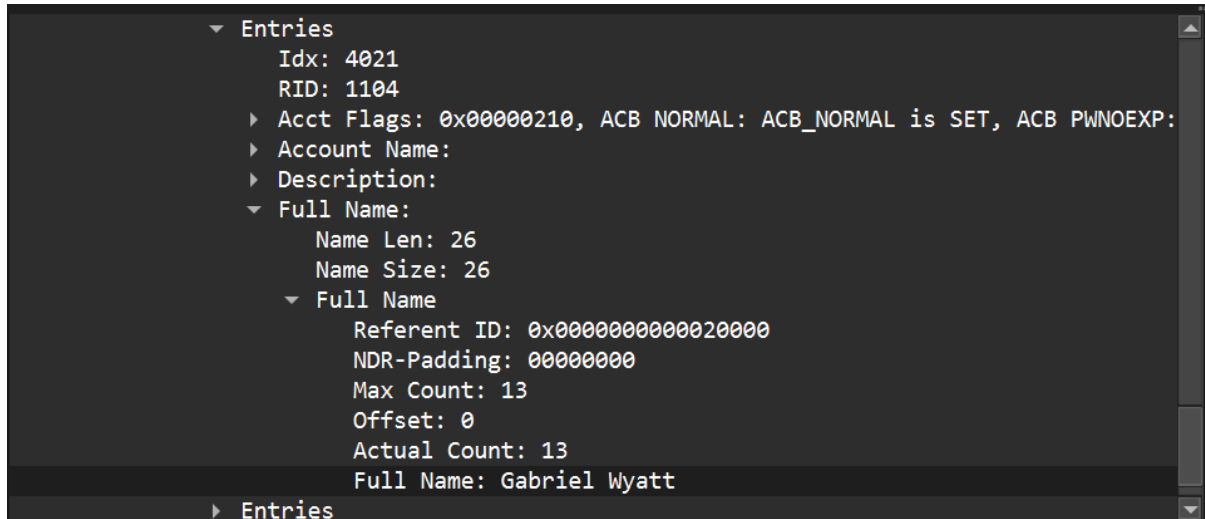
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 - Record Mark: 217 bytes
 - as-req
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 - msg-type: krb-as-req (10)
 - padata: 1 item
 - req-body
 - Padding: 0
 - kdc-options: 40810010
 - cname
 - name-type: KRB5-NT-PRINCIPAL (1)
 - cname-string: 1 item
 - CNameString: gwyatt

User Account Name: **gwyatt**

What is the full name of the user from the user account?

To obtain more details about the user, the Find Packet feature in Wireshark was used to search for occurrences of the username “gwyatt”

Among the results, we found some additional user information. Inspecting the packet revealed a field labelled Full Name, which identified the user as:



Full Name: **Gabriel Wyatt**

What is the domain from 153.92.1[.]49 that triggered the alert for Lumma Stealer?

By using the filter,

“(http.request) and ip.addr eq 153.92.1.49”

to filter out http packet and involving the malicious IP that triggered the alert, we will see the request packets. By having the Host as a new column we can see the domain from which the alert for Lumma Stealer triggered.

The image shows a Wireshark packet list pane with the filter 'ip.addr == 153.92.1.49'. The table displays the following data:

No.	Time	Source	Source Port	Destination	Host	Destination Port	Protocol	Length	Info
25298	103.780107	10.1.21.58	50943	153.92.1.49	whitepepper.su	80	HTTP	558	GET /favicon.ico HTTP/1.1
25286	103.554215	10.1.21.58	49295	153.92.1.49	whitepepper.su	80	HTTP	1149	POST /api/set_agent?id=3BF67EC05320C5729578BE
25270	103.190587	10.1.21.58	49295	153.92.1.49	whitepepper.su	80	HTTP	615	GET /api/set_agent?id=3BF67EC05320C5729578BE4
24614	96.485831	10.1.21.58	59097	153.92.1.49	whitepepper.su	80	HTTP	546	GET /favicon.ico HTTP/1.1
24601	96.287894	10.1.21.58	54492	153.92.1.49	whitepepper.su	80	HTTP	1197	POST /api/set_agent?id=3BF67EC05320C5729578BE
24500	95.344550	10.1.21.58	54492	153.92.1.49	whitepepper.su	80	HTTP	603	GET /api/set_agent?id=3BF67EC05320C5729578BE4

Domain: **whitepepper[.]su**

Summary of Findings

Artifact	Value
Infected Host IP	10.1.21[.]58
MAC Address	00:21:5d:c8:0e:f2
Hostname	DESKTOP-ES9F3ML
User Account	gwyatt
Full Name	Gabriel Wyatt
Malicious Domain	whitepepper[.]su
Malicious IP	153.92.1[.]49

Conclusion

Through systematic analysis of network traffic using Wireshark, we successfully identified the compromised Windows system and associated user details. By leveraging protocol specific filters such as IP, DHCP, and Kerberos, and http we were able to pivot from a known malicious IP to uncover critical host-level indicators and network indicators which triggered the alert. This investigation highlights the importance of network visibility and structured analysis techniques in incident response and threat hunting.
